

RESEARCH REPORT SUMMARY

PROJECT: APPLICATION OF ARTIFICIAL INTELLIGENCE TO MITIGATE CREDIT BARRIERS FOR SMALL AND MEDIUM ENTERPRISES (SMEs) IN VIETNAM

PART 1. INTRODUCTION

1. Rationale and Urgency of the Topic

Small and Medium Enterprises (SMEs) constitute a dominant proportion of the Vietnamese corporate landscape, playing a pivotal role in creating job opportunities, driving structural economic growth, and fostering cross-sector innovation. Nevertheless, the capacity of these enterprises to secure official institutional credit lines remains profoundly restricted due to a constellation of systemic bottlenecks: a chronic shortage of high-quality tangible collateral, inadequate levels of financial transparency, unstandardized internal accounting metrics, and conventional credit underwriting frameworks that remain heavily dependent on manual appraisal methodologies. Consequently, commercial banks encounter extreme operational difficulty in measuring and pricing credit risk accurately.

In the contemporary era of nationwide digital transformation and the exponential advancement of Artificial Intelligence (AI), engineering financial risk assessment architectures grounded in AI and machine learning algorithms presents a robust paradigm shift. This technological intervention yields major practical dividends: enhancing predictive underwriting accuracy, drastically minimizing overhead appraisal expenditures, expanding clean capital access corridors for underserved SMEs, and equipping credit institutions with the data tools required to finalize rapid, optimization-driven lending decisions. Therefore, conducting deep academic and practical research into this domain carries substantial theoretical merit and immense socio-economic urgency.

2. Research Objectives

This research seeks to systematically harmonize the existing theoretical frameworks governing SME credit constraints, institutional credit risk definitions, and modern AI/Machine Learning deployment models within the banking sector. By conducting an empirical, data-driven analysis of the current structural impediments impeding SME financing across Vietnam, the project isolates the core financial determinants that directly amplify credit vulnerability. Ultimately, the study develops and proposes a specialized AI-driven risk management model tailored to the idiosyncratic operational features of Vietnamese commercial banks, systematically mitigating credit rationing boundaries for the SME sector.

3. Central Research Questions

1. What constitute the primary structural and behavioral barriers that artificially restrict SME credit access within the official Vietnamese financial landscape?
2. Which quantitative financial metrics exert the most statistically significant impact on the shifting probability of credit default among domestic SMEs?

3. Through what specific mechanisms can AI and advanced machine learning algorithms optimize predictive accuracy and neutralize the vulnerabilities of traditional credit scoring?
4. Does the integration of alternative data streams into an AI framework yield a measurable expansion of capital access boundaries for unrated or collateral-deficient SMEs?

4. Scope and Methodology

Research Scope: The empirical focus encompasses small and medium enterprises operating across major economic zones in Vietnam, paired with lending portfolios and underwriting standards of commercial banking institutions over a designated historical timeline spanning from 2020 to 2025.

Research Methodology: The study synthesizes a dual qualitative-quantitative framework. The qualitative component centers on exhaustive document tracking, comparative regulatory analysis, and policy framework synthesis. The quantitative engine relies on descriptive statistics, multi-variable logistic regression modeling, and advanced ensemble machine learning algorithms (specifically Random Forest and Extreme Gradient Boosting - XGBoost).

PART 2. CORE RESEARCH CONTENT & THEORETICAL FRAMEWORK

I. Literature Review and Foundational Theories

1. SMEs and Institutional Credit Barriers

Under the statutory framework of Decree 80/2021/NĐ-CP in Vietnam, Small and Medium Enterprises are strictly classified based on distinct thresholds regarding total active labor forces, annual revenue aggregates, and total asset capital capitalization, tailored across three macro-sectors: Agriculture-Forestry-Fisheries, Industry-Construction, and Commerce-Services. Collectively, SMEs represent approximately 98% of all registered operating enterprises in Vietnam, generating roughly 40% of national GDP, absorbing over 50% of the active labor pool, and accounting for nearly 30% of state budget revenues (VCCI, 2023). This highlights their undeniable role as the backbone of domestic economic vitality.

In economic theory, the phenomenon of a credit constraint refers to a market friction where an enterprise is completely denied credit lines or receives capital allocations substantially below its legitimate operational requirements despite presenting economically viable initiatives. This failure originates fundamentally from the theory of information asymmetry between lenders and borrowers (Stiglitz & Weiss, 1981), inducing adverse selection and moral hazard. In Vietnam, this structural asymmetry causes commercial banks to enforce rigid credit rationing, driving disproportionately high loan rejection frequencies and onerous, collateral-heavy borrowing preconditions.

2. Credit Risk and Underwriting Dimensions

According to the standard codifications issued by the Basel Committee on Banking Supervision (BCBS), credit risk is defined as the probability that an economic counterparty will fail to fulfill its committed financial obligations in accordance with the pre-agreed terms of a credit contract, inflicting a direct capital write-down on the lender. From a regulatory perspective (IMF, 2006; State Bank of Vietnam Circular 11/2021/TT-NHNN), an asset is downgraded to a Non-Performing Loan (NPL) or bad debt when the corresponding principal or

interest obligations are past due for 90 days or longer, or when objective indicators reveal severe impairment in the debtor's repayment capacity. Legally, these are categorized into Group 3 (Substandard), Group 4 (Doubtful), and Group 5 (Loss).

To measure and control this threat, traditional systems rely on qualitative evaluation matrices, primarily the 5Cs framework (Character, Capacity, Capital, Collateral, Conditions) or classical quantitative statistical architectures like the Altman Z-Score model for bankruptcy prediction:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

Where zones are demarcated into Safe ($Z > 2.99$), Grey ($1.81 \leq Z \leq 2.99$), and Distress ($Z < 1.81$). However, these formulas fail to handle massive, non-linear alternative data inputs, highlighting the necessity for adaptive AI algorithms.

II. Empirical Analysis of SME Credit Profiles in Vietnam

To quantify the financial drivers behind institutional loan rejections, the research team isolated five typical Vietnamese SMEs representing key operational sectors, tracking their core financial performance over the 2020–2025 cycle.

Enterprise Profile	Annual Revenue (VND B)	ROA (%)	ROE (%)	Debt / Total Assets (%)	Debt / Equity (%)	Net Profit (VND B)
Vinaceglass JSC	12.83	1.75	2.05	14.60	17.00	1.70
Binh Duong Packaging Co.	4.38	13.41	14.21	6.11	6.51	4.92
Central Pharma JSC 3 (TW3)	65.46	5.56	11.05	69.58	228.70	5.82
Food Products JSC	26.94	4.50	4.66	4.45	4.66	4.72
Water Supply JSC	85.21	0.16	0.25	33.35	50.04	4.50

Note: Red values indicate high risk based on traditional bank underwriting standards (ROA/ROE < 2%; Debt/Equity > 100%). Source: Audited Financial Reports.

1. Granular Corporate Profile Diagnostic

- **CTCP Vinaceglass:** Exhibits structural cash flow volatility. Historical tracking reveals a collapse in revenue from 71.92 billion VND (2020) down to 17.92 billion VND (2021)—a 75% contraction in a single fiscal year —pushing ROA and ROE into deeply negative territory. This lack of cash flow stability makes future repayment capacity unpredictable for traditional underwriting models.

- **Binh Duong Packaging Co.:** Possesses an exceptionally healthy capital structure with minimal debt leverage (Debt/Equity of 6.51%) and high profitability metrics. However, its small revenue base (4.38 billion VND) automatically triggers rigid asset-scale thresholds in commercial banks, disqualifying it from accessing substantial capital expansions.
- **Dược Trung Ương 3 (TW3):** Generates robust revenue volumes but maintains an aggressive debt position. Its Debt/Equity ratio reaches a dangerous 228.70%, exceeding the standard financial safety threshold by 2.3 times. This extreme leverage amplifies default vulnerability if interest rates fluctuate upward.
- **Water Supply JSC:** Demonstrates a classic corporate paradox. Despite capturing the largest revenue footprint (85.21 billion VND), its operating efficiency is near-zero (ROA: 0.16%, ROE: 0.25%). Because the enterprise generates virtually no return on assets, it forms a high-risk credit profile incapable of maintaining structural debt service coverage.

2. The Triple Bottleneck Conclusion

The empirical diagnostic proves that SME credit constraints in Vietnam are driven by a interconnected "Triple Bottleneck": (1) Weak asset profitability (ROA/ROE below threshold metrics), elevating bank risk perceptions; (2) Distorted capital leverage structures (Debt/Equity far exceeding safety zones); and (3) Aggressive operational cash flow volatility. When these three factors converge, traditional credit evaluation architectures trigger automatic rejections, trapping SMEs in a credit starvation loop.

PART 3. PROPOSED AI MODEL ARCHITECTURE & SWOT ASSESSMENT

To overcome these structural bottlenecks, the study proposes integrating advanced ensemble Machine Learning architectures—specifically fusing **XGBoost (Extreme Gradient Boosting)** and **Random Forest** algorithms—into the commercial bank underwriting process.

STRENGTHS (S)	WEAKNESSES (W)
• Unrivaled capacity to process massive, multi-dimensional, and unstructured alternative data streams. • Iterative boosting mechanisms (XGBoost) naturally detect hidden, non-linear relationships and risk variables. • Delivers significantly higher predictive precision (Area Under Curve - AUC) compared to classical linear regressions.	• The "Black-Box" Effect: Highly complex algorithmic decisions are inherently difficult to interpret transparently relative to simple logistic regressions. • Absence of Linear Transparency: Lacks intuitive, fixed criteria weighting models, complicating immediate visual audit tracking for compliance officers.
OPPORTUNITIES (O)	THREATS (T)
• Breaking the Credit Loop: Shifts credit scoring emphasis away from heavy asset collateral toward dynamic real-time cash flow vectors. • Personalized Risk Pricing: Computes highly precise Probability of Default (PD) values, allowing banks to offer competitive, tiered interest rates. • Automation Scaling: Enables Straight-Through Processing (STP) for automated loan approvals, lowering operational expenses (OPEX).	• Data Discontinuity: The model depends heavily on massive, clean data inputs; however, many Vietnamese SMEs lack comprehensive digital accounting records. • High Initial CAPEX: Demands extensive upfront investments in secure data warehousing infrastructures and specialized data engineering personnel. • Macroeconomic Volatility: Systemic external economic shocks can cause algorithmic drift, requiring continuous model retraining.

III. Strategy for Algorithmic Risk Management

1. **Neutralizing the Black-Box Vulnerability via XAI:** To fulfill the regulatory transparency requirements enforced by the State Bank of Vietnam, banking platforms must deploy Explainable AI (XAI) frameworks, such as SHAP (SHapley Additive exPlanations) vectors. This technology disassembles the ensemble matrix output, translating mathematical decisions into transparent, auditable rationales by explicitly listing the top financial variables driving the credit approval or rejection.
2. **Data Cleanliness and the "Human-in-the-Loop" Shield:** To neutralize input data deficiencies and structural biases, commercial banks must enforce a rigorous "Human-in-the-Loop" architecture. The machine learning engine acts as an automated, high-speed scoring filter, but the final authorization for edge cases, borderline scores, or complex SME corporate restructurings remains subject to manual review by senior credit underwriters.

PART 4. POLICY RECOMMENDATIONS & FUTURE ARCHITECTURE

1. Operational Re-engineering for Commercial Banks

Commercial banking networks must systematically transition from traditional, static asset-collateral underwriting frameworks to dynamic, transaction-driven, alternative data-based credit underwriting models. By linking banking application programming interfaces (APIs) directly into digital accounting software ecosystems, corporate e-invoice networks, and real-time electronic tax transaction systems, banks can gather a continuous stream of operational cash flow data. This data provides an authentic view of an enterprise's current economic health, effectively bypassing the informational blind spots found in historical balance sheets.

2. Macro-Level Government Policy Interventions

To optimize credit delivery channels, state authorities should execute three coordinated interventions: First, structurally expand the capital reserves and expand the statutory operational mandate of local Credit Guarantee Funds, replicating successful international models like South Korea's KODIT framework to insulate commercial lenders against unsecured default exposures. Second, replace rigid, uniform interest rate frameworks with risk-stratified interest rate subsidy matrices that adjust dynamically to an enterprise's operational scale. Third, launch national financial literacy initiatives to assist micro-enterprise owners in digitizing operations and formalizing accounting records to meet modern underwriting standards.

3. Future Development: The Multi-Layered Financial Ecosystem

Looking toward long-term transformation, the focus must shift from simply expanding bank credit to engineering a sophisticated, multi-layered financial ecosystem for SMEs. This includes building clear regulatory pathways for small-scale corporate bond markets, specialized venture capital funds, and micro-financing P2P lending platforms. For SMEs, enhancing internal accounting governance and transparency must be recognized as a core strategic capability. In financial markets, structural transparency operates as an essential market signal (Spence, 1973)—a fundamental signal that builds long-term institutional trust, lowers capital acquisition costs, and secures sustainable corporate expansion.

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